JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO GUNA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

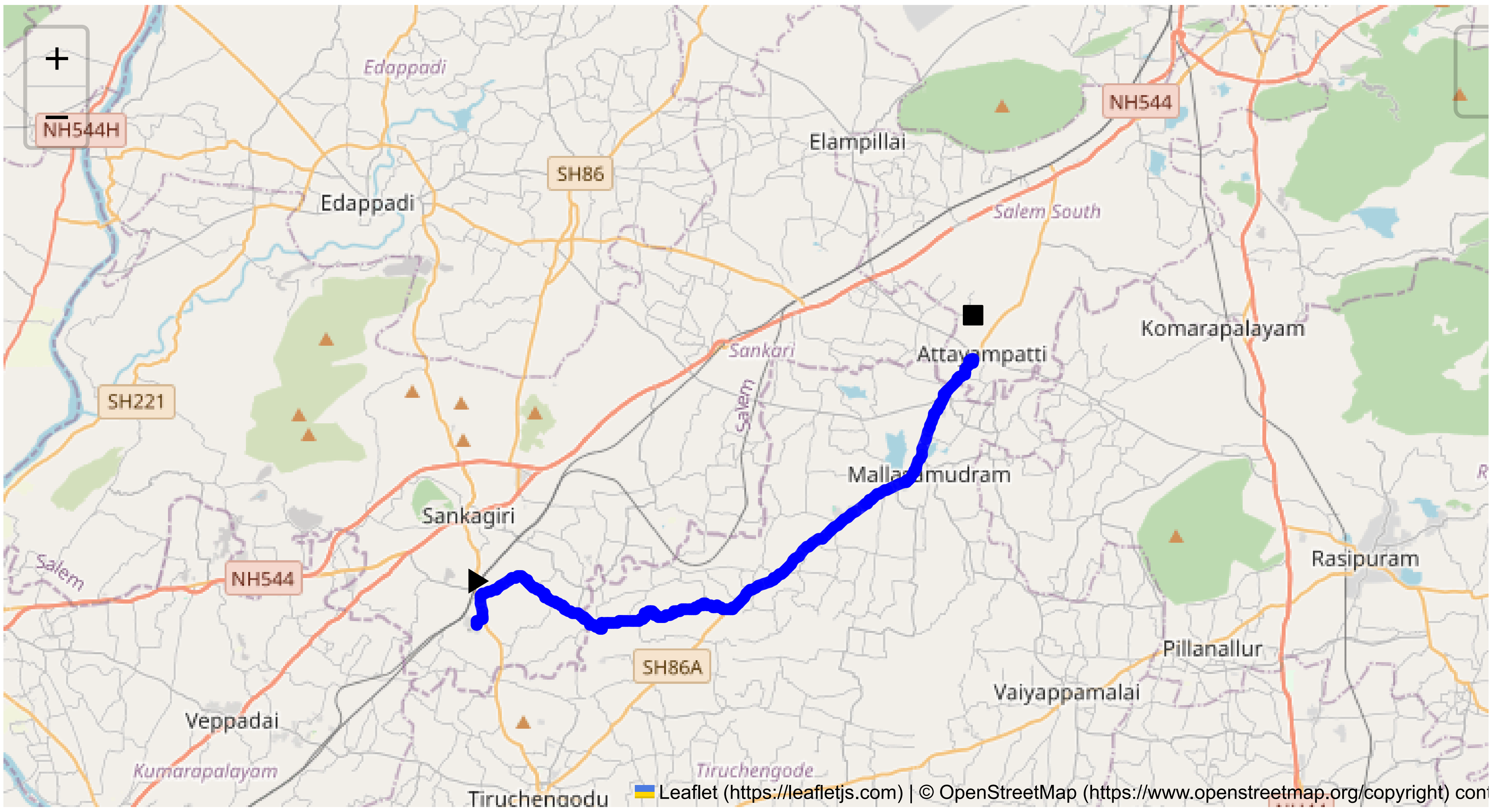
Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

Route Summary:
Total Distance: 26.73 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.529821, 78.046806)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Attayampatti, passing through Puthur, Sankari R.S Rd, Pillanatham, and Mallasamudram, covers approximately 26.73 kilometers. It primarily traverses rural areas, including local roads and main roads such as the Tiruchengode - Salem Main Rd.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** Tamil Nadu generally experiences tropical climate conditions with high temperatures and heavy monsoonal rains.
- **Monsoon Hazards:** From June to September, the southwest monsoon could lead to heavy rainfall, resulting in poor visibility and road flooding.
- **Heat:** High temperatures can lead to road surface damage or exacerbate vehicle-related issues.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic tends to peak during the morning (8-10 AM) and evening (5-7 PM) as these are the commuting hours for local residents.
- **Congestion Areas:** Sangagiri and Attayampatti may experience congestion due to local market activities and junctions where local roads meet major highways.

4. Assessment of Road Quality and Infrastructure

- **Road Conditions:** The road quality varies, with some sections well-paved, while others may have potholes or uneven surfaces, particularly after the monsoon season.
- **Infrastructure:** Signage and road demarcations might be inadequate in rural stretches, and some narrow bridges or intersections require careful navigation.

5. Suggestions for Alternative Routes for Emergencies

In case of blockages or roadworks, consider route diversions through other nearby local roads, possibly bypassing some village roads by joining larger highways earlier, if accessible.

6. Summary of Local Regulations Affecting Hazardous Material Transport

The transportation of hazardous materials requires compliance with regulations established by the Central Motor Vehicle Rules in India, including appropriate vehicular signage, driver certification, and emergency equipment documentation.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

The region has had instances of road accidents involving heavy vehicles, often attributed to inadequate vehicle maintenance or driver fatigue. However, specific incidents involving hazardous materials are less documented but require caution due to rural road constraints.

8. Environmental Considerations and Sensitive Areas

- **Wildlife Crossings:** The route may feature occasional wildlife crossings, especially near agricultural lands, demanding caution.
- **Water Bodies:** Proximity to small bodies of water might raise concerns about potential contamination during spills, particularly during monsoonal runoffs.

9. Analysis of Communication Coverage

Telecommunications coverage is moderate, with potential dead zones in more rural sections or valleys between hills, impacting mobile network reliability.

10. Estimated Emergency Response Times for Different Route Segments

Emergency services might take longer to reach rural parts, with response times estimated at 30-45 minutes depending on the proximity to urban centers.

12. An Overall Summary of Risk Assessment

This route presents several risks typical of rural Indian roads, such as inconsistent road quality, variable traffic congestion, and potential communication and emergency service delays. Weather conditions, especially monsoons, significantly impact travel safety. Ensuring vehicle maintenance, route familiarity, and adherence to transport regulations are crucial for minimizing risks while navigating this path with hazardous materials.

Overall, this route requires careful planning and preparedness for unexpected conditions, prioritizing safety for both the driver and surrounding communities.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43968, 77.87345	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44861, 77.87429	15 KM/Hr
5	Blind Spot	Blind Spot	11.43673, 77.91703	10 KM/Hr
6	Turn	High	11.43926, 77.91705	15 KM/Hr
7	Turn	Medium	11.43936, 77.92301	30 KM/Hr
8	Turn	Medium	11.43997, 77.92587	30 KM/Hr
9	Turn	Medium	11.44030, 77.93066	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.44075, 77.93216	30 KM/Hr
11	Turn	Medium	11.44088, 77.93223	30 KM/Hr
12	Turn	Medium	11.44127, 77.93218	30 KM/Hr
13	Turn	Medium	11.44243, 77.93244	30 KM/Hr
14	Turn	Medium	11.44254, 77.93256	30 KM/Hr
15	Turn	Medium	11.44172, 77.94140	30 KM/Hr
16	Turn	Medium	11.44369, 77.96036	30 KM/Hr
17	Turn	High	11.44387, 77.96283	15 KM/Hr
18	Blind Spot	Blind Spot	11.52943, 78.04650	10 KM/Hr
19	Turn	Medium	11.52886, 78.04753	30 KM/Hr
20	Blind Spot	Blind Spot	11.52879, 78.04758	10 KM/Hr
21	Turn	High	11.52958, 78.04783	15 KM/Hr
22	Turn	Medium	11.52966, 78.04777	30 KM/Hr
23	Turn	High	11.52972, 78.04738	15 KM/Hr
24	Turn	High	11.52968, 78.04735	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	police	Mallasamudram Police Station	11.4998427, 78.0303028	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium
4	school	Mahendra Matriculation School	11.5120354, 78.0354797	30 km/h	Medium
5	college	Mahendra Arts and science college	11.5112305, 78.0344287	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44861, 77.87429

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.43673, 77.91703

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43926, 77.91705

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43936, 77.92301

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43997, 77.92587

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44030, 77.93066

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44075, 77.93216

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44088, 77.93223

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44127, 77.93218

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44243, 77.93244

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44254, 77.93256

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44172, 77.94140

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44369, 77.96036

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44387, 77.96283

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.52943, 78.04650